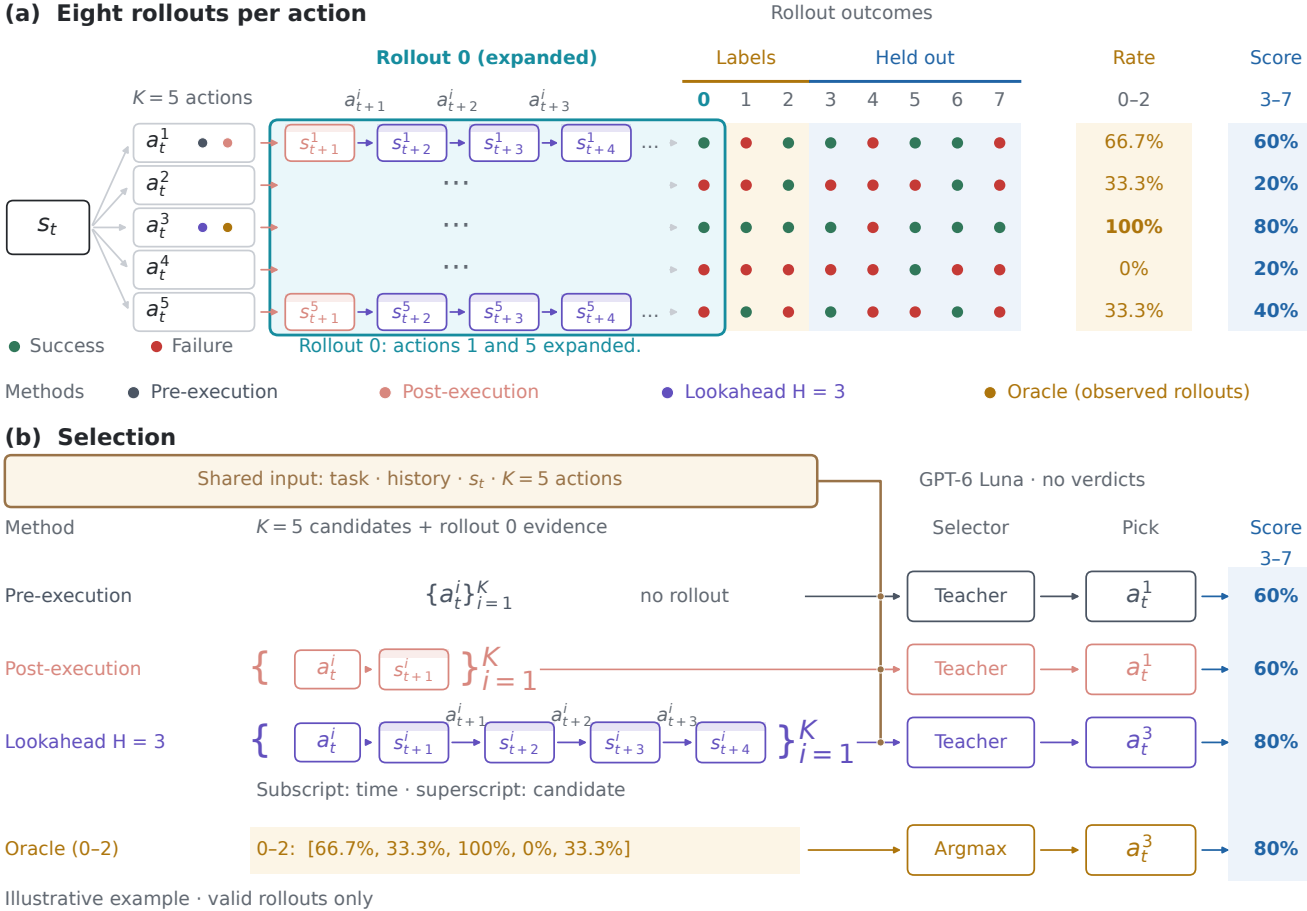
(a) Eight rollouts per action
Rollout outcomes
Rollout 0 (expanded)
Labels
Held out
Rate
Score
K = 5 actions
0 1 2 3 4 5 6 7
0–2
3–7
s_t
a_t^1 a_t^2 a_t^3 a_t^4 a_t^5
a_{t+1}^i a_{t+2}^i a_{t+3}^i
s_{t+1}^1 s_{t+2}^1 s_{t+3}^1 s_{t+4}^1
s_{t+1}^5 s_{t+2}^5 s_{t+3}^5 s_{t+4}^5
66.7%
33.3%
100%
0%
33.3%
60%
20%
80%
20%
40%
Success
Failure
Rollout 0: actions 1 and 5 expanded.
Methods
Pre-execution
Post-execution
Lookahead H = 3
Oracle (observed rollouts)
(b) Selection
Shared input: task · history · s_t · K = 5 actions
GPT-6 Luna · no verdicts
Method
K = 5 candidates + rollout 0 evidence
Selector
Pick
Score
3–7
Pre-execution
$\{a_t^i\}_{i=1}^K$
no rollout
Teacher
a_t^1
60%
Post-execution
$\{ a_t^i \rightarrow s_{t+1}^i \}_{i=1}^K$
Teacher
a_t^1
60%
Lookahead H = 3
$\{ a_t^i \rightarrow s_{t+1}^i \xrightarrow{a_{t+1}^i} s_{t+2}^i \xrightarrow{a_{t+2}^i} s_{t+3}^i \xrightarrow{a_{t+3}^i} s_{t+4}^i \}_{i=1}^K$
Teacher
a_t^3
80%
Subscript: time · superscript: candidate
Oracle (0–2)
0–2: [66.7%, 33.3%, 100%, 0%, 33.3%]
Argmax
a_t^3
80%
Illustrative example · valid rollouts only